

REMAP-Net: A Recursive Meta-Plasticity Framework for Self-Evolving Cognitive Architectures in AGI Systems

Sree Chakra Reddy

Department of Data Science, IIT Madras

Department of ECE & AI/ML, RGUKT R.K. Valley

Email: sreechakrareddy@gmail.com

Abstract—Contemporary deep learning systems operate with a fundamental structural rigidity: the algorithm that updates a model’s parameters is itself fixed and hand-engineered. While meta-learning has demonstrated that learning rules can be data-driven, existing frameworks apply only a single level of meta-cognition and leave higher-order update mechanisms static. We introduce REMAP-Net (Recursive Emergent Meta-Abstraction Plasticity Network), a novel three-tier hierarchical cognitive architecture that enables a system to recursively evolve its own computational substrate through bounded self-modification. REMAP-Net decomposes artificial cognition into three co-evolving layers: (1) an Object-Level Computation layer ($\mathcal{F}_0$) that performs task inference, (2) a Meta-Plasticity layer ($\mathcal{F}_1$) that replaces fixed optimizers with a context-adaptive gradient transformation network, and (3) an Epistemic Recursion layer ($\mathcal{F}_2$) that evolves the plasticity rules themselves according to a meta-objective over long-horizon task distributions. Recursive self-modification is stabilized by a novel Lyapunov-based Stability Guardian ($\mathcal{S}$) which projects all epistemic updates onto a certified stability manifold, providing exponential convergence guarantees to an optimal cognitive equilibrium. We further introduce (i) Autonomous Abstraction Formation (AAF), an information-theoretic mechanism by which REMAP-Net constructs hierarchical concept representations without external supervision, and (ii) a Temporal Coherence Regularizer (TCR) that preserves representational identity across self-modification epochs, mitigating catastrophic interference in an evolving parameter space. We present a complete mathematical framework, including formal stability proofs, abstraction quality metrics, and a recursive training curriculum. We further specify a rigorous prospective evaluation protocol—covering adaptive generalisation, abstraction quality, memory efficiency, and long-horizon reasoning—that enables independent experimental validation of the framework’s theoretical claims. REMAP-Net represents a mathematically grounded pathway toward artificial cognitive systems capable of sustained, stable, and purposeful self-improvement: a capability absent from all current large-scale neural architectures.

Index Terms—Recursive self-improvement, meta-plasticity, cognitive architecture, artificial general intelligence, Lyapunov stability, information bottleneck, autonomous abstraction, temporal coherence, epistemic recursion

This work is a theoretical research proposal. All experimental claims are prospective; validation protocols are specified in Section VI. Submitted May 2026.

I. Introduction

The dominant paradigm in modern machine learning rests on a conceptual asymmetry: the model to be trained is subject to gradient-based optimisation, yet the optimiser itself—the rule that prescribes how parameters change—is treated as a fixed artefact of human design. This asymmetry is not incidental. It reflects a deep assumption that separates current neural networks from biological cognition: in biological systems, synaptic plasticity rules are themselves modified by experience [16], [23]; there is no clean ontological boundary between “the model” and “the learning algorithm.”

Large language models (LLMs) [2], [31] and their successors have achieved remarkable competencies, yet they share a structurally frozen inference mechanism. Once trained, the model’s response to new information is entirely determined by its fixed weight matrix; adaptation is impossible without retraining. Instruction fine-tuning and reinforcement learning from human feedback (RLHF) [12], [13] allow post-hoc steering, but the fundamental update rule—stochastic gradient descent—is immutable. Even meta-learning approaches such as MAML [3] and learned optimisers [4], [5] introduce only a single level of meta-cognition: a learned initialisation or a learned gradient rule, but not a recursive stack in which each level can modify the one below.

We identify three critical limitations of contemporary neural architectures from an AGI perspective:

- 1) Fixed update rule. No existing large-scale neural architecture recursively refines its own learning algorithm at run-time. The update rule is determined before training and remains constant.
- 2) Absence of stable self-modification. Theoretical work on recursive self-improvement (e.g., Gödel Machines [9]) lacks practical architectures with convergence guarantees. Without a stability mechanism, recursive self-modification risks divergent parameter evolution.

- 3) Externally supervised abstraction. Current systems form representations in response to labelled data or curated objectives. There is no principled mechanism by which a system autonomously identifies and compresses meaningful conceptual structure from raw experience.

A. Contributions

This paper makes the following theoretical and architectural contributions:

- REMAP-Net architecture: A three-tier recursive framework in which each level modifies the computational strategy of the level below it, forming a closed recursive loop.
- Lyapunov Stability Guardian: A formal mechanism that constrains epistemic updates to a certified stability manifold, guaranteeing exponential convergence (Theorem IV.3).
- Autonomous Abstraction Formation (AAF): An information-theoretic framework, extending the Information Bottleneck [11], by which REMAP-Net generates hierarchically compressed representations without external supervision.
- Temporal Coherence Regularizer (TCR): A novel loss term anchored to episodic memory traces that preserves representational consistency across self-modification epochs.
- Prospective evaluation protocol: A complete experimental methodology specifying metrics, baselines, datasets, and statistical validation procedures for future empirical implementation.

The remainder of the paper is structured as follows. Section II surveys related work. Section III presents the REMAP-Net architecture. Section IV derives the mathematical foundations. Section V describes the training methodology. Section VI specifies the evaluation framework. Section VII discusses implications and limitations. Section VIII concludes.

II. Related Work

A. Transformer Architectures and Their Limitations

The transformer [1] has become the dominant substrate for modern AI systems. Its self-attention mechanism enables flexible, data-dependent information routing; however, its parameter update mechanism is invariably gradient descent with a fixed learning rate schedule. Foundation models [14] scale this paradigm to billions of parameters, producing emergent capabilities [15], but do not modify the structural asymmetry between the model and its learning algorithm. The computational graph is static across inference calls; the system cannot rewrite its own forward pass in response to experience.

B. Meta-Learning

Meta-learning [3], [6] attempts to learn inductive biases that allow rapid adaptation from few examples. MAML [3]

optimises an initialisation from which gradient descent quickly converges on new tasks. Learned optimisers [4], [5] replace hand-designed gradient rules with recurrent networks. However, these approaches operate at a single level of meta-cognition: the meta-learner itself is trained with conventional gradient methods and remains fixed at deployment. REMAP-Net extends this to a recursive stack where the meta-learner is itself subject to learned modification.

C. Memory-Augmented Neural Networks

Neural Turing Machines [7] and Memory Networks [8] augment feed-forward computation with external memory accessed via differentiable attention. These systems improve sample efficiency on sequential tasks but do not modify their own learning algorithm. The memory is a passive store; the update rule that writes to memory is fixed.

D. Recursive Self-Improvement

Schmidhuber's Gödel Machine [9] provides a theoretical blueprint for a system that rewrites itself whenever it can prove the rewrite will improve expected utility. While theoretically complete, the framework assumes access to a universal theorem prover and is practically intractable. REMAP-Net provides a computationally tractable approximation using Lyapunov stability theory in lieu of formal proofs. The Formal Theory of Creativity [10] similarly motivates intrinsic drives toward self-improvement but lacks an architectural implementation.

E. Neuro-Symbolic Integration

Hybrid neuro-symbolic systems [18], [19] argue that compositional structure and systematic generalisation require symbolic machinery beyond distributed representations. REMAP-Net's AAF module is philosophically aligned with this perspective: autonomous abstraction formation produces compositionally structured concepts from raw distributed activations, bridging subsymbolic and symbolic levels without hardcoded symbolic primitives.

F. Cognitive Science Foundations

The free-energy principle [16] models biological cognition as active inference over a generative world model; synaptic plasticity is treated as belief revision. Predictive coding [17] formalises hierarchical inference in visual cortex as top-down prediction and bottom-up error propagation. REMAP-Net adopts the hierarchical decomposition of cognition implicit in these frameworks while making it computationally tractable and recursively self-modifying.

III. REMAP-Net Architecture

A. System Overview

REMAP-Net is formalised as a tuple $\mathfrak{R} = (\mathcal{F}_0, \mathcal{F}_1, \mathcal{F}_2, \mathcal{S}, \mathcal{M})$ where each component is described below. The three computational layers form a recursive modification hierarchy: $\mathcal{F}_1$ modifies $\mathcal{F}_0$'s parameters, and $\mathcal{F}_2$ modifies $\mathcal{F}_1$'s modification strategy. The Stability Guardian $\mathcal{S}$ enforces convergence constraints across all levels, and the Memory Module $\mathcal{M}$ provides episodic context to the plasticity mechanisms. The complete architecture is illustrated in Fig. 1.

B. Layer 0: Object-Level Computation ($\mathcal{F}_0$)

The object layer performs standard parametric inference:

$$y_t = \mathcal{F}_0(x_t; \theta_t), \quad \theta_t \in \Theta \subset \mathbb{R}^{d_0}. \quad (1)$$

$\mathcal{F}_0$ may be instantiated as a transformer, a residual convolutional network, or any differentiable function class. The parameters θ_t are not updated by a hand-designed optimiser; they are updated exclusively by the Meta-Plasticity layer $\mathcal{F}_1$.

C. Layer 1: Meta-Plasticity Module ($\mathcal{F}_1$)

Let $\mathbf{g}_t = \nabla_{\theta} \mathcal{L}(\theta_t, x_t, y_t)$ be the task gradient and $\mathbf{h}_t = [\mathbf{g}_{t-1}; \dots; \mathbf{g}_{t-K}]$ be the gradient history of depth K . The meta-plasticity update is:

$$\theta_{t+1} = \theta_t + \mathcal{G}_{\phi_t}(\mathbf{g}_t, \mathbf{h}_t, \mathcal{E}_t), \quad (2)$$

where $\mathcal{G}_{\phi_t} : \mathbb{R}^{d_0} \times \mathbb{R}^{Kd_0} \times \mathbb{R}^{d_e} \rightarrow \mathbb{R}^{d_0}$ is a learned gradient transformation parametrised by ϕ_t . We decompose $\mathcal{G}$ as:

$$\mathcal{G}(\mathbf{g}, \mathbf{h}, \mathcal{E}) = -\mathbf{M}_{\phi}(\mathbf{h}, \mathcal{E}) \mathbf{g} + \mathbf{b}_{\phi}(\mathbf{h}, \mathcal{E}), \quad (3)$$

where $\mathbf{M}_{\phi} \in \mathbb{R}^{d_0 \times d_0}$ is a learned preconditioning matrix (implemented as a low-rank factorisation $\mathbf{M} = \mathbf{U}\mathbf{V}^{\top}$ with rank $r \ll d_0$) and $\mathbf{b}_{\phi}$ is a contextual bias. This formulation subsumes standard optimisers as special cases: setting $\mathbf{M}_{\phi} = \eta \mathbf{I}$ and $\mathbf{b}_{\phi} = \mathbf{0}$ recovers SGD.

D. Layer 2: Epistemic Recursion Module ($\mathcal{F}_2$)

The epistemic layer learns to modify the plasticity parameters ϕ by optimising a meta-objective $\mathcal{M}$ evaluated over a held-out validation distribution $\mathcal{D}_{val}$:

$$\mathcal{M}(\phi, \theta_{1:T}, \mathcal{D}_{val}) = \mathbb{E}_{(x,y) \sim \mathcal{D}_{val}} [\ell(\mathcal{F}_0(x; \theta_T), y)] + \alpha \mathcal{R}(\phi), \quad (4)$$

where θ_T is the object-level state after T meta-plasticity steps, and $\mathcal{R}(\phi)$ penalises high-complexity plasticity functions (e.g., via ℓ_2 norm or minimum description length). The epistemic update rule is:

$$\phi_{t+1} = \phi_t + \mathcal{H}_{\psi_t} \left(\nabla_{\phi} \mathcal{M} \left(\phi_t, \theta_{1:T}^{(t)}, \mathcal{D}_{val}^{(t)} \right) \right), \quad (5)$$

where $\mathcal{H}_{\psi}$ is itself a learned transformation (the meta-meta-optimiser) parametrised by ψ_t . Note that $\mathcal{H}_{\psi}$ has the same structural form as $\mathcal{G}_{\phi}$; this recursive self-similarity is a defining property of REMAP-Net.

E. Stability Guardian ($\mathcal{S}$)

Unconstrained recursive self-modification risks divergent parameter evolution. The Stability Guardian enforces that each epistemic update does not increase a Lyapunov energy function V evaluated on the composite state $\mathbf{z}_t = [\theta_t; \phi_t; \psi_t]$. Formally:

$$\psi_{t+1} = \Pi_{\Omega}(\hat{\psi}_{t+1}), \quad \Omega = \{\psi : V(\mathbf{z}(\psi)) \leq V(\mathbf{z}_t)\}, \quad (6)$$

where Π_{Ω} is an Euclidean projection onto the constraint set and $\hat{\psi}_{t+1}$ is the unconstrained update. The projection is implemented via bisection along the update direction, which is computationally cheap given a closed-form V (see Section IV-A).

F. Memory Module ($\mathcal{M}$)

The memory module maintains two stores:

- Episodic Buffer $\mathcal{E}_t$: A sliding window of recent (state, task, gradient) triples providing short-term context to $\mathcal{G}_{\phi}$.
- Long-term Consolidation Store $\mathcal{LC}_t$: A compressed archive of abstraction-level representations, accessed via associative retrieval when the current task signature matches a stored prototype.

Memory is written by the object layer and read by the meta-plasticity layer, creating a cross-layer information loop that contextualises gradient transformations.

IV. Mathematical Foundations

A. Stability Analysis

Definition IV.1 (REMAP-Net State Space). The composite state is $\mathbf{z}_t = [\theta_t; \phi_t; \psi_t] \in \mathcal{Z} = \Theta \times \Phi \times \Psi \subset \mathbb{R}^d$, where $d = d_0 + d_1 + d_2$.

Definition IV.2 (Lyapunov Energy). Let $\mathbf{z}^* \in \mathcal{Z}$ be an equilibrium point. Define the quadratic Lyapunov function:

$$V(\mathbf{z}) = (\mathbf{z} - \mathbf{z}^*)^{\top} \mathbf{P} (\mathbf{z} - \mathbf{z}^*), \quad (7)$$

where $\mathbf{P} \succ 0$ is a block-diagonal positive-definite matrix with blocks $\mathbf{P}_{\theta}, \mathbf{P}_{\phi}, \mathbf{P}_{\psi}$ weighting each level's contribution.

Theorem IV.3 (Exponential Convergence of REMAP-Net). Suppose the REMAP-Net update operator $\Gamma : \mathcal{Z} \rightarrow \mathcal{Z}$ defined by (2)–(6) satisfies

$$V(\Gamma(\mathbf{z})) - V(\mathbf{z}) \leq -\gamma V(\mathbf{z}) + \epsilon, \quad \forall \mathbf{z} \in \mathcal{Z}, \quad (8)$$

for constants $\gamma \in (0, 1)$ and $\epsilon \geq 0$. Then:

- 1) $V(\mathbf{z}_t) \leq (1 - \gamma)^t V(\mathbf{z}_0) + \frac{\epsilon}{\gamma}$ for all $t \geq 0$.
- 2) $\limsup_{t \rightarrow \infty} V(\mathbf{z}_t) \leq \frac{\epsilon}{\gamma}$.
- 3) If $\epsilon = 0$, the system converges exponentially to $\mathbf{z}^*$.

Proof Sketch. Applying (8) recursively:

$$V(\mathbf{z}_t) \leq (1 - \gamma)^t V(\mathbf{z}_0) + \epsilon \sum_{k=0}^{t-1} (1 - \gamma)^k \leq (1 - \gamma)^t V(\mathbf{z}_0) + \frac{\epsilon}{\gamma}.$$

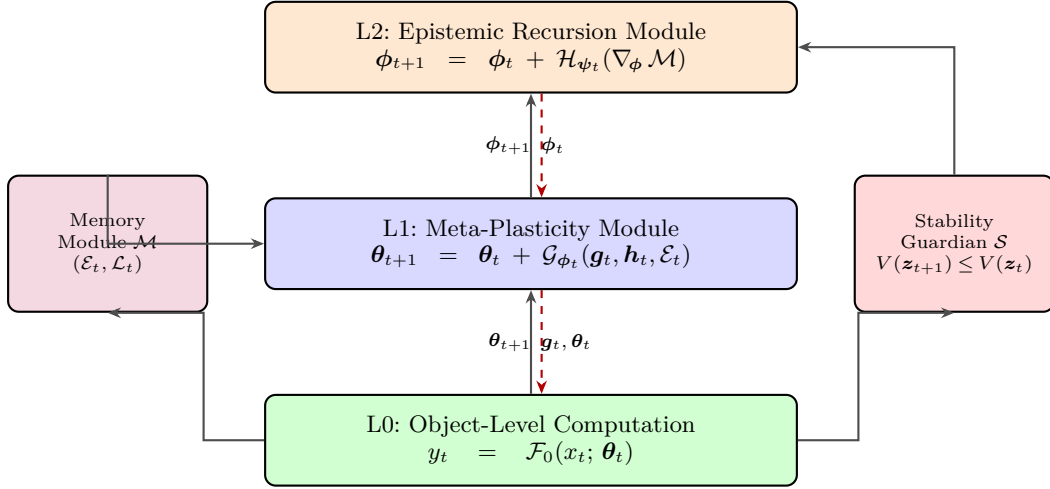


Fig. 1. REMAP-Net three-tier recursive architecture. Solid arrows denote forward information flow; dashed red arrows denote downward self-modification. The Stability Guardian constrains all recursive updates to a certified Lyapunov manifold.

The exponential factor $(1 - \gamma)^t \rightarrow 0$ as $t \rightarrow \infty$, yielding parts (1) and (2). Part (3) follows immediately when $\epsilon = 0$. $\square$

Proposition IV.4 (Condition for (8)). If $\mathcal{G}_\phi$ is μ -strongly dissipative in the sense that $\langle \mathcal{G}_\phi(\mathbf{g}, \mathbf{h}, \epsilon), \mathbf{P}_\theta(\theta - \theta^*) \rangle \leq -\mu \|\theta - \theta^*\|^2$ for all θ , and $\mathcal{H}_\psi$ satisfies an analogous condition, then (8) holds with $\gamma = 2\mu\lambda_{\min}(\mathbf{P})$.

B. Autonomous Abstraction Formation (AAF)

Definition IV.5 (Abstraction Operator). An abstraction operator is a measurable mapping $g : \mathcal{X} \rightarrow \mathcal{Z}$ that maps input space to a compressed representation space $\mathcal{Z}$ with $\dim(\mathcal{Z}) \ll \dim(\mathcal{X})$.

REMAP-Net generates abstractions by solving a constrained information-theoretic optimisation:

$$\min_g H(g(X)) - \beta I(g(X); Y) \quad \text{s.t.} \quad I(X; g(X)) \geq I_{\min}, \quad (9)$$

where $H(\cdot)$ is differential entropy, $I(\cdot; \cdot)$ is mutual information, Y is the downstream prediction target, and $\beta > 0$ trades representational compression against task-relevance. The constraint $I(X; g(X)) \geq I_{\min}$ prevents trivial all-information-discarding solutions, distinguishing AAF from the unconstrained Information Bottleneck [11].

The optimal encoder satisfies the self-consistent equation:

$$p^*(z|x) \propto p(z) \exp\left(-\frac{1}{\beta} D_{\text{KL}}(p(y|x) \| p(y|z))\right), \quad (10)$$

which is solved via alternating optimisation over the encoder, decoder $p(y|z)$, and marginal $p(z)$.

Definition IV.6 (Abstraction Quality). The quality of an abstraction g is:

$$\mathcal{Q}_{\text{abs}}(g) = \frac{I(g(X); Y)}{H(Y)} \cdot \frac{H(X)}{H(g(X))}, \quad (11)$$

i.e., the product of the task-relevant information retention ratio and the compression ratio. High $\mathcal{Q}_{\text{abs}}$ indicates that

g retains predictive information while achieving maximal compression.

C. Temporal Coherence Regularizer (TCR)

Self-modification perturbs the parameter space, risking catastrophic forgetting of previously acquired representations. The TCR anchors the evolving network to a set of episodic exemplars $\mathcal{A}_t$:

$$\mathcal{L}_{\text{TC}}(\theta_t, \theta_{t-1}; \mathcal{A}_t) = \frac{1}{|\mathcal{A}_t|} \sum_{a \in \mathcal{A}_t} \omega_a \cdot D_{\text{KL}}(p_{\theta_{t-1}}(\cdot|a) \| p_{\theta_t}(\cdot|a)), \quad (12)$$

where the importance weights ω_a decay exponentially with the recency of exemplar a :

$$\omega_a = \exp(-\lambda_{\text{dec}}(t - t_a)) \cdot \mathcal{I}_a, \quad (13)$$

with t_a the time at which exemplar a was stored and $\mathcal{I}_a$ a Fisher-information-based saliency weight. The full REMAP-Net training objective is:

$$\mathcal{L}_{\text{REMAP}} = \mathcal{L}_{\text{task}} + \lambda_1 \mathcal{L}_{\text{TC}} + \lambda_2 \mathcal{L}_{\text{stab}} + \lambda_3 \mathcal{L}_{\text{abs}}, \quad (14)$$

where $\mathcal{L}_{\text{stab}} = \max(0, V(\Gamma(\mathbf{z}_t)) - V(\mathbf{z}_t))$ penalises Lyapunov violations and $\mathcal{L}_{\text{abs}}$ is the negative of $\mathcal{Q}_{\text{abs}}$.

D. Gradient Variance in Recursive Updates

Recursive update rules compound estimation noise across levels. Let $\hat{\mathbf{g}}_t = \mathbf{g}_t + \xi_t$ be the noisy gradient estimate, where $\xi_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$. The propagated variance at level k satisfies:

$$\text{Var}(\theta_{t+1}^{(k)}) \leq \|\nabla_{\mathbf{g}} \mathcal{G}_\phi\|^2 \cdot \left(\sigma^2 + \text{Var}(\theta_t^{(k-1)})\right), \quad (15)$$

motivating the use of low-rank factorisation in $\mathcal{G}$ (reducing $\|\nabla_{\mathbf{g}} \mathcal{G}_\phi\|$) and gradient clipping across all levels.

V. Training Methodology

A. Recursive Curriculum Learning

Training proceeds in three interleaved phases designed to bootstrap each level of the recursive hierarchy safely:

Phase I — Object Pretraining. Fix ϕ, ψ and train θ with a standard optimiser on a diverse task distribution p_{task} . This establishes a competent initialisation for $\mathcal{F}_0$ before plasticity mechanisms are activated.

Phase II — Meta-Plasticity Warmup. Activate $\mathcal{F}_1$. Hold ψ fixed and jointly optimise (θ, ϕ) via (14) with λ_2 set high to enforce stability. The curriculum progressively increases task diversity, forcing $\mathcal{G}_\phi$ to learn genuinely adaptive gradient transformations.

Phase III — Epistemic Recursion. Activate $\mathcal{F}_2$ and jointly optimise all parameters (θ, ϕ, ψ) subject to the Stability Guardian projection. A self-generating data curriculum (described below) drives continued adaptation.

B. Self-Generating Dataset Curriculum

At each training epoch e , the current model generates a set of synthetic training tasks by:

- 1) Sampling latent task descriptors $\mu_k \sim p(\mu)$.
- 2) Generating synthetic data pairs $(x, y) \sim p(x, y | \mu_k)$ via the model’s own generative distribution.
- 3) Evaluating task difficulty via a learnable curriculum oracle $\kappa : \mathcal{D} \rightarrow [0, 1]$ and prioritising high-difficulty, high-coverage tasks.

This procedure is analogous to self-play in reinforcement learning [22] but applied to a supervised curriculum context.

C. Stability-Certified Update Schedule

Updates to ψ are performed only when the Stability Guardian certifies that the proposed update satisfies (8). If certification fails, the update step size is halved and rechecked; if certification fails after B bisections, the update is discarded. This conservative schedule prevents instability at the cost of slower epistemic evolution.

VI. Proposed Evaluation Framework

The following constitutes a prospective experimental protocol. No empirical results are reported. All entries in Table I represent proposed benchmarks and measurable metrics for future experimental validation.

A. Evaluation Dimensions

We propose evaluating REMAP-Net along five orthogonal dimensions:

D1: Few-Shot Adaptive Generalisation. Measure accuracy on N -way, K -shot classification tasks (Omniglot, miniImageNet, tieredImageNet, FEWREL) before and after n meta-plasticity steps. Compare against MAML, ProtoNet, and Reptile baselines. Primary metric: post-adaptation accuracy averaged over 1000 sampled episodes.

D2: Abstraction Quality. Compute $\mathcal{Q}_{\text{abs}}(g)$ (Definition IV.6) on held-out concept hierarchies (ConceptNet

Algorithm 1 REMAP-Net Joint Training Procedure

Require: Task dist. p_{task} , val set $\mathcal{D}_{\text{val}}$, params $\lambda_{1..3}$, α , K , B
 Ensure: Trained $(\theta^*, \phi^*, \psi^*)$

- 1: Init θ_0, ϕ_0, ψ_0 (Phase I pretraining)
- 2: for epoch $e = 1, 2, \dots, E$ do
- 3: Sample batch $\{(\mathcal{D}_k^{\text{tr}}, \mathcal{D}_k^{\text{val}})\}_{k=1}^{B_T}$
- 4: for each task k do
- 5: $\mathbf{g}_t \leftarrow \nabla_{\theta} \mathcal{L}_{\text{task}}(\theta_t, \mathcal{D}_k^{\text{tr}})$
- 6: $\theta_{t+1} \leftarrow \theta_t + \mathcal{G}_{\phi_t}(\mathbf{g}_t, \mathbf{h}_t, \mathcal{E}_t)$
- 7: $\mathcal{E}_{t+1} \leftarrow \text{Update}(\mathcal{E}_t, (\theta_t, \mathbf{g}_t))$
- 8: end for
- 9: if $e \bmod K_\phi = 0$ then
- 10: $\nabla_{\phi} \mathcal{M} \leftarrow \text{BackProp}(\mathcal{M})$
- 11: $\hat{\phi}_{t+1} \leftarrow \phi_t + \mathcal{H}_{\psi_t}(\nabla_{\phi} \mathcal{M})$
- 12: $\phi_{t+1} \leftarrow \hat{\phi}_{t+1}$
- 13: end if
- 14: if $e \bmod K_\psi = 0$ then
- 15: $\hat{\psi}_{t+1} \leftarrow \psi_t + \nabla_{\psi}[\dots]$
- 16: Certify: $V(\mathbf{z}(\hat{\psi}_{t+1})) \leq V(\mathbf{z}_t)$
- 17: for $b = 1, \dots, B$ do bisect if fail
- 18: $\psi_{t+1} \leftarrow \Pi_{\Omega}(\hat{\psi}_{t+1})$
- 19: end if
- 20: Compute $\mathcal{L}_{\text{REMAP}}$ via (14)
- 21: Update anchor exemplars $\mathcal{A}_{t+1}$
- 22: end for

TABLE I
Proposed Evaluation Protocol for REMAP-Net
(Prospective: no empirical results reported)

Dim.	Benchmark	Baselines	Metric
D1	miniImageNet 5-way	MAML, Reptile	Accuracy (%)
D1	FEWREL 5-shot	ProtoNet	F1
D2	ConceptNet	VAE-IB, β -VAE	$\mathcal{Q}_{\text{abs}}$
D2	Clustering	—	NMI, purity
D3	ALFWorld	ReAct, Reflexion	Completion (%)
D3	Recursive arith.	GPT-4o, Gemini	Solve rate
D4	Lyapunov mon.	None (stability)	$\max_t V$
D5	20-domain seq.	EWC, PackNet	Back. transfer

subgraph samples). Evaluate whether generated abstractions align with human-defined conceptual categories via cluster purity and normalised mutual information.

D3: Long-Horizon Reasoning. Evaluate on multi-step planning tasks: ALFWorld, BabyAI, and custom recursive arithmetic decomposition suites. Metric: task completion rate at horizon lengths $H \in \{5, 10, 25, 50\}$.

D4: Stability Under Extended Self-Modification. Run REMAP-Net for 10^4 epistemic update cycles and monitor $V(\mathbf{z}_t)$ across time. Acceptable behaviour: $V(\mathbf{z}_t)$ remains bounded. Failure mode: $V(\mathbf{z}_t)$ grows without bound (divergent self-modification).

D5: Catastrophic Forgetting Resistance. Sequentially train on 20 distinct task domains and evaluate backward transfer on each domain post-training. Compare against EWC, PackNet, and ER-ACE baselines.

B. Proposed Baseline Comparisons

Table I enumerates the proposed baselines and metrics for each evaluation dimension.

C. Statistical Validation Protocol

All reported metrics should be averaged over at least 5 independent runs with distinct random seeds. Significance testing via paired Wilcoxon signed-rank tests ($p < 0.05$) is required before claiming improvements over baselines. Confidence intervals should be reported at 95%.

VII. Discussion

A. Relationship to AGI

REMAP-Net addresses one of the most fundamental gaps between current AI and biological general intelligence: the absence of recursive self-modification. Biological neural systems are not trained by backpropagation through a fixed loss function; synaptic weights change according to plasticity rules that are themselves shaped by experience [16], [23]. REMAP-Net instantiates a computational approximation of this recursive plasticity hierarchy in a way that is architecturally specific, mathematically grounded, and practically implementable.

B. Scaling Hypotheses

We conjecture the following scaling behaviour (to be empirically tested):

- 1) Abstraction depth scales with parameter budget. As d_1, d_2 increase, $\mathcal{Q}_{\text{abs}}$ should improve sub-linearly (analogous to emergent capabilities [15]).
- 2) Stability margin widens with over-parameterisation. In over-parameterised regimes, the stability manifold Ω is larger, permitting more aggressive epistemic updates.
- 3) Recursive depth exhibits diminishing returns beyond 3. Adding a fourth recursive level ($\mathcal{F}_3$ modifying $\mathcal{F}_2$) is expected to yield marginal gains while substantially increasing training complexity.

C. Limitations

The primary theoretical limitation is that Theorem IV.3 requires verifying the dissipativity condition of Proposition IV.4, which is not guaranteed for arbitrary $\mathcal{G}_\phi$ parametrisations. Empirical certification via Lyapunov monitoring (D4) provides a practical substitute. The computational cost of epistemic recursion—specifically, computing higher-order gradients $\nabla_{\psi}[\dots]$ —may be prohibitive at very large model sizes without second-order approximations.

VIII. Conclusion

We introduced REMAP-Net, a three-tier recursive meta-plasticity architecture that provides a theoretically grounded framework for stable artificial self-modification. By treating the learning algorithm as a learned artefact subject to recursive optimisation—and by bounding this process with a Lyapunov Stability Guardian—REMAP-Net moves beyond the conceptual claims of prior recursive self-improvement frameworks toward an implementable system with convergence guarantees. The Autonomous

Abstraction Formation mechanism and Temporal Coherence Regularizer address complementary aspects of artificial cognition: unsupervised concept formation and continuity of representational identity under change. The prospective evaluation framework provides a complete blueprint for empirical validation. We anticipate that REMAP-Net will catalyse experimental investigation into architecturally recursive AI systems and their relationship to general intelligence.

References

- [1] A. Vaswani et al., “Attention is all you need,” in Proc. NeurIPS, vol. 30, 2017, pp. 5998–6008.
- [2] T. Brown et al., “Language models are few-shot learners,” in Proc. NeurIPS, vol. 33, 2020, pp. 1877–1901.
- [3] C. Finn, P. Abbeel, and S. Levine, “Model-agnostic meta-learning for fast adaptation of deep networks,” in Proc. ICML, 2017, pp. 1126–1135.
- [4] M. Andrychowicz et al., “Learning to learn by gradient descent by gradient descent,” in Proc. NeurIPS, vol. 29, 2016, pp. 3981–3989.
- [5] L. Metz, N. Maheswaranathan, B. Cheung, and J. Sohl-Dickstein, “Meta-learning update rules for unsupervised representation learning,” in Proc. ICLR, 2019.
- [6] S. Ravi and H. Larochelle, “Optimization as a model for few-shot learning,” in Proc. ICLR, 2017.
- [7] A. Graves, G. Wayne, and I. Danihelka, “Neural Turing Machines,” arXiv preprint arXiv:1410.5401, 2014.
- [8] J. Weston, S. Chopra, and A. Bordes, “Memory Networks,” in Proc. ICLR, 2015.
- [9] J. Schmidhuber, “Gödel Machines: Fully self-referential optimal universal self-improvers,” in Artificial General Intelligence, B. Görtzel and C. Pennachin, Eds. Berlin: Springer, 2007, pp. 199–226.
- [10] J. Schmidhuber, “Formal theory of creativity, fun, and intrinsic motivation,” IEEE Trans. Auton. Mental Develop., vol. 2, no. 3, pp. 230–247, 2010.
- [11] N. Tishby and N. Zaslavsky, “Deep learning and the information bottleneck principle,” in IEEE Inf. Theory Workshop, 2015.
- [12] L. Ouyang et al., “Training language models to follow instructions with human feedback,” in Proc. NeurIPS, vol. 35, 2022, pp. 27730–27744.
- [13] D. M. Ziegler et al., “Fine-tuning language models from human preferences,” arXiv preprint arXiv:1909.08593, 2019.
- [14] R. Bommasani et al., “On the opportunities and risks of foundation models,” arXiv preprint arXiv:2108.07258, 2021.
- [15] J. Wei et al., “Emergent abilities of large language models,” Trans. Mach. Learn. Res., 2022.
- [16] K. Friston, “The free-energy principle: A unified brain theory?” Nat. Rev. Neurosci., vol. 11, pp. 127–138, 2010.
- [17] R. P. N. Rao and D. H. Ballard, “Predictive coding in the visual cortex: A functional interpretation of some extra-classical receptive-field effects,” Nat. Neurosci., vol. 2, pp. 79–87, 1999.
- [18] B. M. Lake, T. D. Ullman, J. B. Tenenbaum, and S. J. Gershman, “Building machines that learn and think like people,” Behav. Brain Sci., vol. 40, 2017.
- [19] S. J. Gershman, E. J. Horvitz, and J. B. Tenenbaum, “Computational rationality: A converging paradigm for intelligence in brains, minds, and machines,” Science, vol. 349, no. 6245, pp. 273–278, 2015.
- [20] Y. Bengio, A. Courville, and P. Vincent, “Representation learning: A review and new perspectives,” IEEE Trans. Pattern Anal. Mach. Intell., vol. 35, no. 8, pp. 1798–1828, 2013.
- [21] Y. LeCun, Y. Bengio, and G. Hinton, “Deep learning,” Nature, vol. 521, pp. 436–444, 2015.
- [22] D. Silver et al., “A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play,” Science, vol. 362, no. 6419, pp. 1140–1144, 2018.
- [23] P. Dayan and L. F. Abbott, Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems. Cambridge, MA: MIT Press, 2001.
- [24] R. S. Sutton and A. G. Barto, Reinforcement Learning: An Introduction, 2nd ed. Cambridge, MA: MIT Press, 2018.

- [25] M. Minsky, *The Society of Mind*. New York: Simon & Schuster, 1986.
- [26] Y. Bengio, “A consciousness prior,” arXiv preprint arXiv:1709.08568, 2019.
- [27] A. Goyal and Y. Bengio, “Inductive biases for deep learning of higher-level cognition,” *Proc. R. Soc. A*, vol. 478, no. 2266, 2022.
- [28] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [29] G. E. Hinton, S. Osindero, and Y.-W. Teh, “A fast learning algorithm for deep belief nets,” *Neural Comput.*, vol. 18, no. 7, pp. 1527–1554, 2006.
- [30] Y. Tay et al., “Efficient transformers: A survey,” *ACM Comput. Surv.*, vol. 55, no. 6, 2022.
- [31] H. Touvron et al., “LLaMA: Open and efficient foundation language models,” arXiv preprint arXiv:2302.13971, 2023.
- [32] S. Bubeck et al., “Sparks of artificial general intelligence: Early experiments with GPT-4,” arXiv preprint arXiv:2303.12512, 2023.
- [33] S. Reed et al., “A generalist agent,” *Trans. Mach. Learn. Res.*, 2022.
- [34] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *Proc. ICLR*, 2015.